॥वसुधैव कुटुम्बकम्॥

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 22

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – Sem III
Title of Assignment	Demonstration and exploration of JDBC classes and interfaces (PreparedStatement and ResultSet).
Practice questions	1. Create a login application using PreparedStatement to check username and password.

```
34 • CREATE TABLE users (username VARCHAR(50), password VARCHAR(50));
35 • INSERT INTO users VALUES ('admin', 'admin123'), ('priya', 'pass@456');
```

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;

public class LoginAppDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String dbUser = "root";
        String dbPassword = "XXXXX";

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter username: ");
        String username = sc.nextLine();
        System.out.print("Enter password: ");
```

```
        String password = sc.nextLine();

        String query = "SELECT * FROM users WHERE username = ? AND
password = ?";

        try (Connection conn = DriverManager.getConnection(url,
dbUser, dbPassword);
             PreparedStatement ps = conn.prepareStatement(query))
{

            ps.setString(1, username);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                System.out.println("Login successful. Welcome, " +
rs.getString("username") + "!");
            } else {
                System.out.println("Login failed. Invalid username
or password.");
            }

        } catch (SQLException e) {
            System.out.println("Database error: " +
e.getMessage());
        }
    }
}
```

2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.



```
CREATE TABLE hospital_staff (login_id VARCHAR(20), password VARCHAR(50), role VARCHAR(20));
INSERT INTO hospital_staff VALUES
('DOC101', 'doc@123', 'Doctor'),
('NUR201', 'nurse@123', 'Nurse');
```

```
import  java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;

public class HospitalLoginDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
```

```
        String dbUser = "root";
        String dbPassword = "XXXXX";

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Login ID: ");
        String loginId = sc.nextLine();
        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT role FROM hospital_staff WHERE
login_id = ? AND password = ?";

        try (Connection conn = DriverManager.getConnection(url,
dbUser, dbPassword);
             PreparedStatement ps = conn.prepareStatement(query))
{

            ps.setString(1, loginId);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                String role = rs.getString("role");
                System.out.println("Access granted. Welcome, " +
role + " (" + loginId + ")");
                if (role.equalsIgnoreCase("Doctor")) {
                    System.out.println("You can view and update
patient treatment records.");
                } else if (role.equalsIgnoreCase("Nurse")) {
                    System.out.println("You can view patient care
schedules.");
                }
            } else {
                System.out.println("Access denied. Invalid Login
ID or password.");
            }

        } catch (SQLException e) {
            System.out.println("Database error: " +
e.getMessage());
        }
    }
}
```

Output	1)
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>java -cp .;mysql`
	`Enter username: admin`
	`Enter password: admin123`
	`Login successful. Welcome, admin!`
	
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>javac -cp .;mysql`
	
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>java -cp .;mysql-`
	`Enter username: priya`
	`Enter password: pass@400`
	`Login failed. Invalid username or password.`
	
	2)
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>java -cp .;mysql`
	`Enter Login ID: DOC101`
	`Enter Password: doc@123`
	`Access granted. Welcome, Doctor (DOC101)`
	`You can view and update patient treatment records.`
	
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>javac -cp .;mysq`
	
	`C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS\JAVA_EXP_22>java -cp .;mysql`
	`Enter Login ID: NUR201`
	`Enter Password: nurse`
	`Access denied. Invalid Login ID or password.`

Conclusion	PreparedStatement precompiles SQL and keeps user input separate from the query, which improves performance and prevents SQL injection. ResultSet then gives sequential access to the retrieved rows. Together they provide the secure, efficient data access that login and authentication systems like these depend on.